

Django Deep Dive: Files & Folders

Lecture #5: Outer vs Inner Directory • manage.py • settings.py • wsgi vs asgi • db.sqlite3

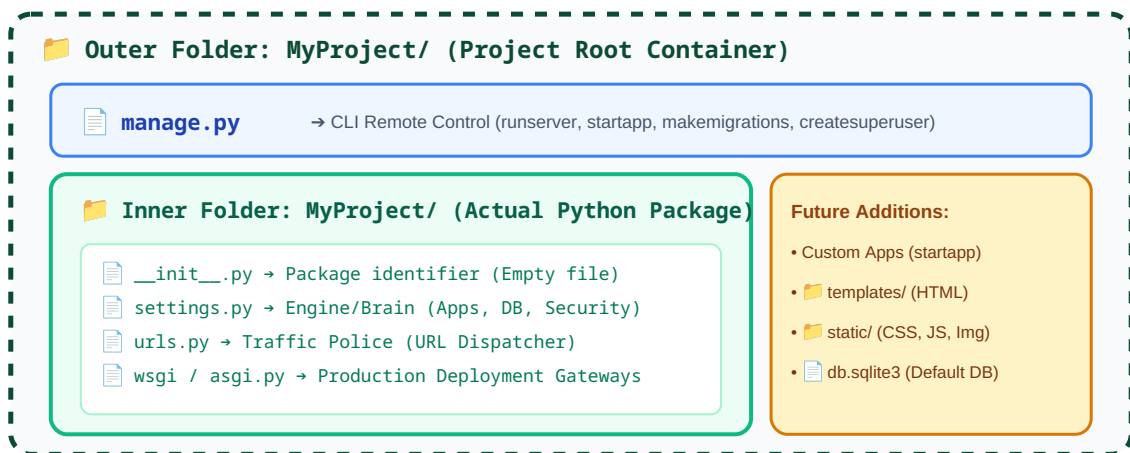
Master Class Notes • Mohit Decodes Hindi Series #5 • Easy Hinglish

🎯 Welcome Soubhik! (Structure & Database Synergy):

Pichle lecture me humne project banaya aur server run kiya tha. Lekin ek professional backend developer banne ke liye har ek file aur folder ka internal mechanism pata hona chahiye! Yahan aapki **SQL & database knowledge** **db.sqlite3** aur **settings.py** ki **DATABASES** config ko samajhne me madad karegi.

🗺️ 1. Outer Folder vs Inner Folder (Visual Architectural Map)

Jab hum `django-admin startproject MyProject` chalte hain, toh do **MyProject** bante hain:



🗂️ 2. Outer Folder vs Inner Folder — Kya Difference Hai?

Directory	Kya Hota Hai Iske Andar?	Asli Role & Responsibility
Outer Folder (MyProject/)	<code>manage.py</code> , aane wali Custom Apps, <code>templates/</code> , <code>static/</code> , aur <code>db.sqlite3</code> .	Project Ka Root Container! Ye pure project ka container box hai. Aap terminal me <code>cd MyProject</code> karke isi folder me kaam karte hain.
Inner Folder (MyProject/)	<code>__init__.py</code> , <code>settings.py</code> , <code>urls.py</code> , <code>wsgi.py</code> , <code>asgi.py</code> .	Actual Project Configuration Package! Ye Python package hai jisme pure project ki global settings aur configuration store hoti hai.

🎮 3. manage.py — Django Ka Master Remote Control

Outer root folder me maujood `manage.py` ek CLI (Command-Line Interface) tool hai. Is file ko kabhi manually touch/edit nahi kiya jata!

Top Tasks Handled by `manage.py` :

- `runserver` : Development server start karta hai (Default Port: 8000).
- `startapp <app_name>` : Project ke andar nayi application/module generate karta hai.
- `makemigrations` & `migrate` : Python models ko SQL tables me translate karke database me apply karta hai.
- `createsuperuser` : Admin panel me login karne ke liye Master Admin account banata hai.

⚙️ 4. Inner Package Files Ka Deep-Dive

File Name	Iske Andar Kya Config Hota Hai?	Real-Life Analogy
<code>settings.py</code>	INSTALLED_APPS : Pre-installed aur custom apps ki list. MIDDLEWARE : Security, sessions, auth filters. DATABASES : SQLite3 (default), Postgres, MySQL config. TEMPLATES : UI HTML files ki directory path. TIME_ZONE , LANGUAGE_CODE , STATIC_URL .	 Company Rulebook & Central Engine : Jahan office ke saare niyam, access keys aur database likhe hote hain.
<code>urls.py</code>	User ke requested URL patterns ko matching View functions ke sath map karta hai (URL Dispatcher). Default me <code>path('admin/', admin.site.urls)</code> rehta hai.	 Receptionist : Visitor ke aate hi usko check karke sahi cabin (View) ka rasta dikhati hai.
<code>__init__.py</code>	Ek empty file jo Python interpreter ko batati hai ki ye folder ek importable Python Package hai.	 Official Stamp : Jo batata hai ki ye aam folder nahi, Python module package hai.
<code>wsgi.py</code> vs <code>asgi.py</code>	WSGI : <i>Web Server Gateway Interface</i> (Traditional synchronous servers like Nginx + Gunicorn). ASGI : <i>Asynchronous Server Gateway Interface</i> (Modern async servers, WebSockets, real-time chats).	 Main Entry Gates : Ek traditional entry gate hai aur dusra modern high-speed express entry gate!

⚡ 5. Server Run Karne Ke 3 Advance Tarike & Cache Files

Lecture #5 me sir ne local server ko alag-alag requirements ke hisaab se run karne ke 3 power methods sikhayee hain:

Tarika 1: Default Port 8000

```
python manage.py runserver
```

Default `http://127.0.0.1:8000/` par start hota hai.

Tarika 2: Custom Port (e.g. 8080)

```
python manage.py runserver 8080
```

Agar port 8000 busy ho toh `http://127.0.0.1:8080/` par chalayein.

Tarika 3: Wi-Fi / Local Network Sharing

```
python manage.py runserver 0.0.0.0:8000
```

Apne laptop ke Wi-Fi IP par start karein. Same Wi-Fi se connected Mobile phone ya kisi dusre PC par direct site live test kar sakte hain bina code move kiye!

Automatic Generated Files

- __pycache__** : Python compiled bytecode cache jo agali baar server ko super-fast boot karta hai (Production me nahi jaata).
- db.sqlite3** : Django ka default lightweight relational database jo server start hote hi ready ho jata hai.

💡 6. Top 5 Interview / Viva Quick Revision Q&A

1. **Q: Django project ke Outer aur Inner folder me kya primary difference hai?**

Ans: Outer folder project ka *Root Container* hota hai jisme `manage.py`, custom apps, aur database file hoti hai; jabki Inner folder actual *Configuration Package* hota hai jisme `settings.py` aur `urls.py` hoti hain.

2. **Q: `manage.py` ka primary use kya hota hai?**

Ans: Ye command-line management utility hai jisse local server chalana (`runserver`), database migrate karna (`migrate`), aur apps banana sambhav hota hai.

3. **Q: `settings.py` me kaun-kaun se core configurations rehte hain?**

Ans: `INSTALLED_APPS`, `MIDDLEWARE`, `DATABASES`, `TEMPLATES`, aur timezone/localization settings.

4. **Q: WSGI aur ASGI me kya farq hai?**

Ans: **WSGI** traditional synchronous HTTP requests handle karta hai, jabki **ASGI** asynchronous requests, WebSockets aur real-time applications ke liye modern standard hai.

5. **Q: Mobile par same Wi-Fi ke andar Django app ko test karne ke liye server kaise chalte hain?**

Ans: `python manage.py runserver 0.0.0.0:8000` command chalakar phone ke browser me system ka Wi-Fi IP address hit karte hain.